

UTI Phenotype Labeling

Clinician Reviewer Instructions

Last Updated: 27 January 2024

Stanford

UTI Phenotype Labeling

Introduction

- Thank you for participating in the review process for our UTI phenotyping project
- This study aims to refine large language models (LLMs) for more accurate and clinically meaningful identification of urinary tract infections (UTIs) within electronic health records (EHRs)
- Phenotyping in this context refers to the systematic classification of patient cases based on clinical data, ensuring that the LLM learns patterns that align with real-world diagnostic reasoning

UTI Phenotype Labeling

Introduction

- As a reviewer, you will be presented with a series of cases, each consisting of a medical record number (MRN) and corresponding urine culture results
- Your task is to identify and review the associated clinical notes to determine which contain meaningful data that can be used to fine-tune the LLM for UTI classification
- Additionally, we ask that you provide insights into the clinical assessment and diagnostic reasoning for each case
- Your expertise will help ensure that our model captures the nuanced decision-making process involved in diagnosing UTIs, ultimately leading to improved automated phenotyping and clinical decision support

Reviewer Instructions

Select Patient

- On the provided Excel Sheet:
 - Note that your name appears in the first column as the reviewer
 - Copy the patient's MRN and note the urine culture order date and time
- Navigate to EPIC Chart Review:
 - Paste the MRN into the field to open the chart review
 - Do NOT create a new encounter

	A	B	C	D	E	F	
			Urine				Ability
			Culture_Order	Note_			Perce
1	Reviwer_Name	Patient MRN	Date & Time	Date_Time	Note_Type	Note_Author	Symp
2	Dr. Example	1[REDACTED]5	11/11/23 8:48				
3							
4							

EPIC PRODUCTION

Search (Ctrl+Space)

In Basket Schedule Patient Lists MDS Pt Status Chart Prep for Case Telephone Call My Reports Lane Library UpToDate ECG Anywhere Sign My Visits Remind Me Service T

Patient Lists

Edit List Properties Remove Patient Add Patient Wrap Text Copy Paste Patient Report Sign In Sign In Others Sign Out Sign Out Others Navigators Open Chart

Search for a Patient

When creating a new patient, please enter as much demographic data as possible.

Name/MRN SSN Sex Birth date Phone

City State ZIP Transplant Pt

Provider My Patients

Find Patient

Clear

Results Recent Patients

Find Urine Culture

- Navigate to the Labs Section:
 - Scroll to find the patient's urine culture (and likely accompanying urinalysis)
 - *TIP: You can create a filter to display only urine cultures and urinalyses*
- Ensure that the urine culture date and time match EXACTLY what is found on the Excel sheet — **contact us with any inconsistencies**
- *TIP: It is recommended that you review the urinalysis and urine culture results prior to reviewing chart notes*

	A	B	C	D	E	F	
			Urine				Ability
			Culture_Order	Note_			Perce
1	Reviwer_Name	Patient MRN	Date & Time	Date_Time	Note_Type	Note_Author	Symp
2	Dr. Example	1[REDACTED]5	11/11/23 8:48				
3							
4							

The screenshot displays the Epic EHR interface. The top navigation bar includes the 'Labs' tab, which is highlighted with a red box. Below the navigation bar, the '1 Year Ago' section shows a list of results. The 'Urine Culture' result is highlighted with a red box. A red arrow points from this result to the 'Collection Information' panel on the right. The 'Collection Information' panel shows details for the 'Urine Culture' result, including 'Urine, Clean Catch' and '>100,000 CFU/mL Escherichia coli'.

Date/Time	Test
1 Year Ago	
11/11/2023 08:48	Urine Culture
11/11/2023 08:48	Urinalysis Screen, Culture if Positive
2 Years Ago	
3 Years Ago	
5 Years Ago	
10 Years Ago	

Collection Information

Urine
Urine, Clean Catch
Collection
Collected: 11/11/2023 8:48 AM
Received: 11/11/2023 9:15 AM

Urine Culture
Status: Final result Visible to patient: Yes (seen)
Specimen Information: Urine, Clean Catch

1 Result Note
Culture, Urine
Resulting Agency: TVLAB
Susceptibility
>100,000 CFU/mL Escherichia coli !
Escherichia coli
MIC

Identify Useful Notes

- Navigate to Notes/ Trans Section:
 - Scroll to find useful notes that discuss the patient’s symptoms that led to the ordered urine culture
 - E.g., ED Provider Notes, Outpatient Progress Notes, Consultant Notes, etc.
 - In general, notes that are most useful are found in the 24 hours prior to culture, and 48 hours after culture — however, it is up to the individual review to determine utility of note

	A	B	C	D	E	F	
			Urine				Ability
			Culture_Order	Note_			Perce
1	Reviwer_Name	Patient MRN	Date & Time	Date_Time	Note_Type	Note_Author	Symp
2	Dr. Example	1[REDACTED]5	11/11/23 8:48				
3							
4							

MRN:

Adv Care Plan: None

Pref Lab: STANFORD HEALTH CARE TRI-VALLEY LAB [233]

MyHealth: Enrolled

VIDYO DIALER

hover over >

Isolation: None

Coverage: Shca Comm Aet...

Allergies: No Known Allergies

Current Meds: 4

Infection: None

SHC Alliance Patient

Weight:

BMI:

Pulse:

BP:

Chart Review

Labs

ED Visits

Encounters

Notes/Trans

Letters

Anes

Proc

Card

Rad

Meds

Referrals

Media

Other Orders

Episodes

LDAs

Consents

Misc R

Preview

Refresh (2:58 PM)

Review Selected

Route

Image Archive

Edit Note

Rarely Used

Filters

Hide Other Enc

Me

Fellow

Department

Hide Deleted

Procedures

H&P

DC Summary

OP Visit

Notes 2022

Notes 2023

Not all records have been loaded and sorted. Load remaining records to sort all records. Load All Records Hide

Encounter Date	Note Date	Encounter Type	Type	Author	Dept.	Status	Specialty	Fili
02/06/2024	02/06/2024 16:40	Telemedicine	Progress Notes	Kilborn, Debora Corison, NP - Nurse Practi...	FMDV	Signed	Nurse Practitioner	02/
1 Year Ago								
11/13/2023	11/13/2023 17:06	Patient Outreach	Telephone Encounter	Zamora, Rowena - Liaison Case Manage...	TVPCM	Signed		11/
11/13/2023	11/13/2023 17:05	Patient Outreach	Progress Notes	Zamora, Rowena - Liaison Case Manage...	TVPCM	Signed		11/
11/11/2023	11/11/2023 10:40	ED	ED Notes	Lisondra, Angelica, RN - Registered Nurse...	TVPED	Signed		11/
11/11/2023	11/11/2023 10:37	ED	ED Provider Notes	Zubair, Sania, MD - Physician - Emergency	TVPED	Signed	Emergency Medi...	11/
10/25/2023	10/25/2023 15:58	Lab	Progress Notes	Kilborn, Debora Corison, NP - Nurse Practi...	CCLAB	Signed	Nurse Practitioner	10/
10/25/2023	10/25/2023 12:42	Lab	Progress Notes	Kilborn, Debora Corison, NP - Nurse Practi...	CCLAB	Signed	Nurse Practitioner	10/
10/02/2023	10/02/2023 10:00	Office Visit	Clinic Support Note	Agero, Giron, MA - Medical Assistant	FMDV	Signed	Radiology	10/
10/02/2023	10/02/2023 10:00	Office Visit	Proaress Notes	Kilborn, Debora Corison, NP - Nurse Practi...	FMDV	Signed	Nurse Practitioner	10/

Identify Useful Notes

- Identifying and transfer useful note metadata:
 - In this example, only one note (the ED Provider Note) detailed the patient’s urinary symptoms
 - Therefore, transfer this metadata to the Excel Sheet **EXACTLY** as written
 - TIP: You can copy and paste this information directly from the note itself, rather than manually entering the data*
- If there are multiple, useful notes for the accompanying MRN and urine culture, **ADD** additional rows for **EACH** note
 - By default, there are 5 empty rows — add more rows (if needed) or leave extra rows empty*

	A	B	C	D	E	F	G
			Urine Culture_Order	Note_			Ability to Perceive Symptoms
1	Reviwer_Name	Patient MRN	Date & Time	Date_Time	Note_Type	Note_Author	
2	Dr. Example	1[REDACTED]5	11/11/23 8:48	11/11/23 10:37	ED Provider Notes	Zubair, Sania, MD	
3				If there are additional notes, ADD extra rows BELOW the accompanying MRN and Urine Culture Order here...			
4							

The screenshot shows the Epic EHR interface. The 'Chart Review' section is active, and the 'Notes/Trans' tab is selected. A red box highlights the 'Notes/Trans' tab. Another red box highlights a specific note entry for '11/11/2023 10:37' by 'Zubair, Sania, MD - Physician - Emergency'. Red arrows point from the highlighted note entry to the corresponding rows in the Excel sheet above, illustrating the data transfer process.

Encounter Date	Note Date	Encounter Type	Type	Author	Dept.	Status	Specialty	File
02/06/2024	02/06/2024 16:40	Telemedicine	Progress Notes	Kilborn, Debora Corison, NP - Nurse Practi...	FMDV	Signed	Nurse Practitioner	02/
1 Year Ago								
11/13/2023	11/13/2023 17:06	Patient Outreach	Telephone Encounter	Zamora, Rowena Liaison Case Manage...	TVPCM	Signed		11/
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11/11/2023	11/11/2023 10:40	ED	ED Notes	Lisondra, Angelica RN - Registered Nurse...	TVPED	Signed		11/
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10/02/2023	10/02/2023 10:00	Office Visit	Prograss Notes	Kilborn, Debora Corison, NP - Nurse Practi...	FMDV	Signed	Nurse Practitioner	10/

- **Open the note:**
 - Read/ review the note to ensure that it contains urinary symptoms to fine-tune the LLM
- In general, you will answer the questions regarding presence or absence of symptoms, then provide supporting documentation (if applicable)
 - The supporting documentation is **COPIED** and **PASTED** directly from the note!
 - Supporting documentation is **NOT** your own free text!

PRODUCTION -

In Basket
Schedule
Patient Lists
MDS
Pt Station
Chart
Prep for Case
Telephone Call
My Reports
Lane Library
UpToDate
ECG Anywhere
Sign My Visits
Remind M

MRN:

Adv Care Plan: None

Pref Lab: STANFORD HEALTH CARE TRI-VALLEY LAB [233]

MyHealth: Enrolled

VIDYO DIALER

hover over >

Isolation: None

Coverage: Shca Comm Act...

Allergies: No Known Allergies

Current Meds: 4

Infection: None

SHC Alliance Patient

Weight:

BMI:

Pulse:

BP:

Chart Review

Labs
ED Visits
Encounters
Notes/Trans
Letters
Anes
Proc
Card
Rad
Meds
Referrals
Media
Other Orders
Episodes
LDAs
Consents
Misc Rpts

☒ Preview
☒ Refresh (3:29 PM)
☒ Review Selected
☒ Route
☒ Image Archive
☒ Edit Note
☒ Rarely Used

☒ Hide Other Enc
☐ Me
☐ Fellow
☐ Department
☒ Hide Deleted
☐ Procedures
☐ H&P
☐ DC Summary
☐ OP Visit
☐ Notes 2022
☐ Notes 2023

☒ Not all records have been loaded and sorted. Load remaining records to sort all records.

Zubair, Sania, MD
Physician
Emergency

ED Provider Notes
Signed

Date of Service: 11/11/2023 10:37 AM

Signed
Expand All
Collapse All

Stanford

MEDICINE

Health Care

Tri-Valley

Emergency Department

Provider Note

Name:
MRN:

ED Arrival: 11/11/2023 9:34 AM
Room #: HW01

History & Physical

Triage:

Chief Complaint:

Clinical Imp

Acute cystitis

Disposition

☒ Discharge

AVS

- ED After Vi
- ED Additio

Follow-Ups: Foll
(Nurse Practitio

Medication

+ cephalexin 5
+ ondansetron
+ phenazopyri
MEALS

Symptom Glossary

- The following is a list of symptoms to identify in the notes — *note that there are many ways to document the presence or absence of these symptoms; this is what you are training the LLM to identify.*
- You can extract symptoms from ANY and MULTIPLE section(s) of the note (e.g., HPI, ROS, PE, Assessment, etc.)
 - Ability to Perceive Symptoms (*Column G*)
 - Capable of perceiving and expressing symptoms? Patients who are comatose, encephalopathic, or unable to speak may not be able to do so.
 - Dysuria (*Column I*)
 - Pain or discomfort during urination.
 - Subjective Fever (*Column K*)
 - Subjective fever that patient experienced/ is experiencing, but not necessarily febrile during the encounter.
 - Urinary Urgency (*Column M*)
 - Sudden, intense urge to urinate.
 - Urinary Frequency (*Column O*)
 - More frequent urination.
 - Suprapubic Pain/ Tenderness (*Column Q*)
 - Discomfort in the bladder/ lower abdomen
 - Flank (CVA) Pain/ Tenderness (*Column S*)
 - Pain in the back or side near the kidneys.
 - Perineal Pain/ Painful Prostate Exam (*Column U*)
 - Pain in the perineum or during prostate examination.
 - Urinary Incontinence (*Column W*)
 - Loss of bladder control.
 - Macroscopic Hematuria (*Column Y*)
 - Visible blood in the urine.

Accepted Responses

- There are three (3) accepted responses:
 - **Yes** — symptom endorsed by the patient as written in the note
 - *E.g., “Patient endorses dysuria.”*
 - **Negation** — symptom denied by the patient as written in the note
 - *E.g., “Patient denies dysuria.”*
 - **No** — symptom is not mentioned anywhere in the note
 - *It is unknown if the patient endorses or denies the symptom*
- When entering data into the Excel sheet, be sure to include the ENTIRE sentence that contains the symptom (*even if it contains other symptoms*), VERBATIM.

“Yes” Example

- *Yes* — symptom endorsed by the patient as written in the note

History & Physical

Triage:

Chief Complaint

Patient presents with

• Urinary Pain

Pt has been experiencing urgency, lower abd discomfort after urinating x 1 day. Pt took Motrin and 1 pyridium that she had left over.

HPI

54 Y female without any past significant medical history. The patient presents to the ER with complaint of pain, burning with urination. Per the patient, symptom onset in the past 24 hours. The patient works as an ER at Sanford Hospital, reports being fairly busy, has not been drinking a lot of water, and ingesting more coffee than usual. Reports of burning with urination. No suprapubic pressure but there is no lower back pain, nausea, vomiting. Denies any fevers, chills. Has been taking Pyridium remaining from a previous UTI for symptom control. Eating and drinking her baseline. Has irregular menses, has an IUD. Denies positive pregnancy.

Review of Systems pain, burning with urination. Others negative

Physical Exam

Vitals

Oxygen saturation reviewed which is 99% on room air which is normal.

CONSTITUTIONAL: Well-appearing; well-nourished; in no apparent distress.

HEAD: Normocephalic; atraumatic.

EYES: PERRL; EOM intact; conjunctiva and sclera are clear bilaterally.

NOSE, THROAT: no rhinorrhea; normal pharynx with no tonsillar hypertrophy; mucous membranes pink/moist, no erythema, no exudate.

NECK: Supple; non-tender; no cervical lymphadenopathy.

CARD: Normal S1, S2; no murmurs, rubs, or gallops. Regular rate and rhythm.

RESP: Normal respiratory effort; breath sounds clear and equal bilaterally; no wheezes, rhonchi, or rales.

ABD: Normal bowel sounds; non-distended; non-tender; no palpable organomegaly, no masses. No CVA tenderness.

EXT: Normal ROM in all four extremities; non-tender to palpation; distal pulses are normal, no edema.

	I	J
1	Dysuria	Supporting Document for Dysuria
2	Yes	

Patient presents with Urinary Pain. The patient presents to the ER with complaint of pain, burning with urination. Reports of burning with urination. Review of Systems pain, burning with urination.

“Negation” Example

- *Negation* — symptom denied by the patient as written in the note

History & Physical

Triage:

Chief Complaint

Patient presents with

- Urinary Pain

Pt has been experiencing urgency, lower abd discomfort after urinating x 1 day. Pt took Motrin and 1 pyridium that she had left over.

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EXT: Normal ROM in all four extremities; non-tender to palpation; distal pulses are normal, no edema.

	S	T
	Supporting Document for	
	Flank (CVA)	Flank (CVA)
1	Pain/Tenderness	Pain/Tenderness
2	Negation	
	The suprapubic pressure but there is no lower back pain, nausea, vomiting. No CVA tenderness.	

“No” Example

- *No* — symptom is not mentioned anywhere in the note

History & Physical

Triage:

Chief Complaint

Patient presents with

- Urinary Pain

Pt has been experiencing urgency, lower abd discomfort after urinating x 1 day. Pt took Motrin and 1 pyridium that she had left over.

HPI

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EXT: Normal ROM in all four extremities; non-tender to palpation; distal pulses are normal, no edema.

Clinical Decision Rules

ED Course, Data Review & Interpretation:

ED Course as of 11/12/23 1719

Sat Nov 11, 2023

1039 Urinalysis Screen, Culture if Positive(!)

Noted greater than 100 RBCs, 51-100 WBCs, occasional bacteria, otherwise unable to assess nitrites, leuk esterase, the patient has been taking Pyridium. [SZ]

ED Course User Index

[SZ] Zubair, Sania, MD

5:19 PM patient is seen and examined, looks well nontoxic. Comfortable appearing. Does have a UTI based on the UA, and clinical symptoms. Doubt pyelonephritis. Well-appearing. Plan to treat 9 the ER with Keflex, will release home with an Rx of the same for the next 5 days. PCP recheck in 2 days recommended. Agreement, understanding verbalized.

	U	V
1	Perineal Pain/Painful Prostate Exam	Supporting Document for Perineal Pain/Painful Prostate Exam
2	No	

Final Assessment

- Knowing the symptoms/ presentation, lab/ culture results, and patient outcome — the remaining four questions are regarding *your* final assessment of the case:
- UTI?
 - *Did the patient actually have a urinary tract infection?*
- Is any infection present?
 - *Including urinary tract infection, is there evidence of any infection (e.g., pneumonia, cellulitis, etc.)?*
- Presumed infection plausible?
 - *Did the observed symptoms (and urinalysis, if applicable) at the time of urine culture (but not knowing the final urine culture result), suggest that the patient had an infection that needed empiric antibiotics?*
- Prophylactic antibiotics appropriate?
 - *Did the patient require “prophylactic” antibiotics due to one or a combination of the following factors:*
 - Asymptomatic bacteriuria in pregnancy
 - Impending urologic procedure
 - Immunocompromising conditioning (e.g., febrile neutropenia) who may not have expected labs or symptoms

“Final Assessment” Example

	AA	AB	AC	AD	AE	AF	AG	AH
1	UTI?	Any comment or notes to support UTI?	Is any infection present?	Any comment or notes to support Is any infection present?	Presumed Infection Plausible?	Any comment or notes to support Presumed Infection Plausible?	Prophylactic Antibiotics Appropriate?	Any comment or notes to support Prophylactic Antibiotics Appropriate?
2	Yes	Patient endorsed urinary symptoms and had supporting urinalysis (pyuria), with confirmatory urine culture (>100,000 CFU E. coli).	No	UTI is present. No other symptoms to suggest alternative infection.	Yes	Patient had urinary symptoms and supporting urinalysis suggestive of UTI that support the use of empiric antibiotics.	No	Patient is not pregnant, immunocompromised, or have an impending urologic procedure.

Repeat the process for each case

Supplemental Education: Urinary Tract Infections

A reference standard for urinary tract infection research: a multidisciplinary Delphi consensus study

Manu P Bilsen, Simon P Conroy, Caroline Schneeberger, Tamara N Platteel, Cees van Nieuwkoop, Lona Mody, Jeffrey M Caterino, Suzanne E Geerlings, Bela Köves, Florian Wagenlehner, Marleen Kunneman, Leo G Visser, Merel M C Lambregts, on behalf of the UTI Reference Standard Consensus Group*

The absence of a consensus-based reference standard for urinary tract infection (UTI) research adversely affects the internal and external validity of diagnostic and therapeutic studies. This omission hinders the accumulation of evidence for a disease that imposes a substantial burden on patients and society, particularly in an era of increasing antimicrobial resistance. We did a three-round Delphi study involving an international, multidisciplinary panel of UTI experts (n=46) and achieved a high degree of consensus (94%) on the final reference standard. New-onset dysuria, urinary frequency, and urinary urgency were considered major symptoms, and non-specific symptoms in older patients were not deemed indicative of UTI. The reference standard distinguishes between UTI with and without systemic involvement, abandoning the term complicated UTI. Moreover, different levels of pyuria were incorporated in the reference standard, encouraging quantification of pyuria in studies done in all health-care settings. The traditional bacteriuria threshold (10⁵ colony-forming units per mL) was lowered to 10⁴ colony-forming units per mL. This new reference standard can be used for UTI research across many patient populations and has the potential to increase homogeneity between studies.

Introduction

Urinary tract infection (UTI) is one of the most common bacterial infections in the global community.¹ The high incidence and recurrence rate of UTI lead to a decreased quality of life, excessive health-care costs, and substantial use of antimicrobials.^{1,2} UTI diagnosis is commonly based on a combination of symptoms and signs, pyuria, and culture results. Current UTI research primarily focuses on improving diagnostics and developing novel therapeutic and prophylactic modalities, such as new antimicrobials and vaccines.^{3,4} However, UTI studies are impeded by the absence of a consensus-based reference standard for UTI, which has several consequences. First, not having a reference standard introduces bias into estimates of diagnostic accuracy and efficacy (also known as verification bias), affecting the internal validity of a study.⁵ Second, if different criteria are used across studies, results cannot be readily compared, compromising the external validity of a study. These drawbacks are particularly relevant in the context of growing antimicrobial resistance, in which reliable efficacy and safety data on novel antimicrobials for UTI are crucial. Moreover, from an ethical standpoint, it is essential to ensure consistent treatment of study participants and patients, as well as accurate reporting of study findings.

Although several proposed definitions exist, none of these definitions are broadly applicable to the majority of UTI studies. Guidelines were formulated by the US Centers for Disease Control and Prevention primarily for the surveillance of nosocomial and catheter-associated UTI, and the revised McGeer criteria were designed for studies in long-term care facilities, which reduces their applicability in other health-care settings.^{6,7} The European Medicines Agency (EMA) and the US Food and Drug Administration (FDA) have published

guidelines for the development and approval of drugs for the treatment of uncomplicated and complicated UTI, including acute pyelonephritis.^{8–10} However, these guidelines apply different symptom criteria, and definitions of complicated UTI are not uniform. Moreover, the EMA guideline does not specify a minimum number of symptoms, and the FDA guideline does not provide a pyuria threshold for uncomplicated UTI, leaving room for interpretation. Furthermore, it is unclear which research methods were used in the development of these guidelines. Before this consensus study, we did a systematic review evaluating published UTI studies, which showed low adherence to FDA and EMA guidelines.¹¹ Researchers frequently defined UTI based on their own criteria or clinical practice guidelines, leading to heterogeneous UTI definitions across studies. These findings underscore the necessity for a multidisciplinary-supported reference standard for UTI, developed specifically for research purposes. Consequently, the primary aim of this study was to achieve consensus on a reference standard for UTI, applicable to adult women and men, including older patients, who participate in studies focusing on bacterial UTI, excluding those individuals with UTI related to indwelling catheters.

Methods

Study design

To gain consensus on a reference standard, a Delphi study was conducted and reported following recommendations from the guidance on conducting and reporting of Delphi studies (CREDES).¹² The Delphi method has four main characteristics: an expert panel is questioned about the issue of interest; the process is anonymous to reduce the effect of dominant personalities; the questionnaires are iterative in nature;

Lancet Infect Dis 2024; 24: e513–21

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March 5, 2024
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*Members listed in the appendix (pp 21–22)

Department of Infectious Diseases (M P Bilsen MD, Prof L G Visser MD, M M C Lambregts MD) and Medical Decision Making, Department of Biomedical Data Sciences (M Kunneman PhD), Leiden University Medical Center, Leiden, Netherlands; Medical Research Council Unit for Lifelong Health and Ageing, University College London, London, UK (Prof S P Conroy PhD); Center for Infectious Disease Control, National Institute for Public Health and the Environment, Bilthoven, Netherlands (C Schneeberger MD); Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht, Netherlands (T N Platteel MD); Department of Internal Medicine, Haga Teaching Hospital, The Hague, Netherlands (C van Nieuwkoop MD); Department of Public Health and Primary Care, The Hague Health Campus, Leiden University Medical Center, The Hague, Netherlands (C van Nieuwkoop); Division of Geriatric and Palliative Medicine, Department of Internal Medicine, University of Michigan Medical School, Ann Arbor, MI, USA (Prof L Mody MD); Geriatrics Research Education and Clinical Center, VA Ann Arbor Healthcare System, Ann Arbor, MI, USA (Prof L Mody); Department of Emergency Medicine, The Ohio State University, Columbus, OH, USA (Prof J M Caterino MD); Amsterdam UMC, Department of Internal Medicine, Amsterdam Institute for



Systematic review

- 47 UTI studies included
- High heterogeneity of UTI definitions
- Low adherence to previous guidelines

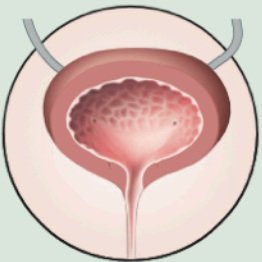


Delphi consensus study

- 57 UTI experts across 11 countries
- Three questionnaire and feedback rounds
- 94% consensus on reference standard

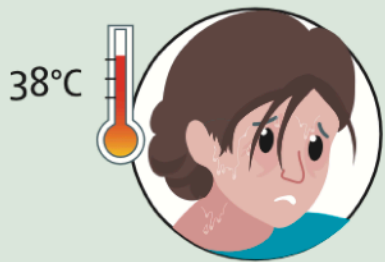


Key recommendations



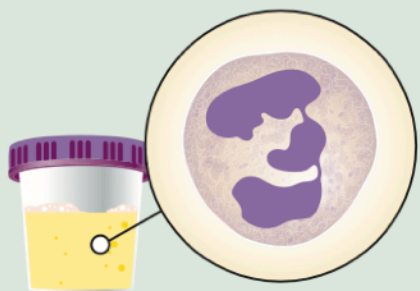
Symptoms and signs

- New-onset dysuria, urinary frequency, and urinary urgency are most indicative of UTI
- Non-specific symptoms should not be included in the definition of UTI
- The reference standard can be applied to patients who are unable to express or experience symptoms



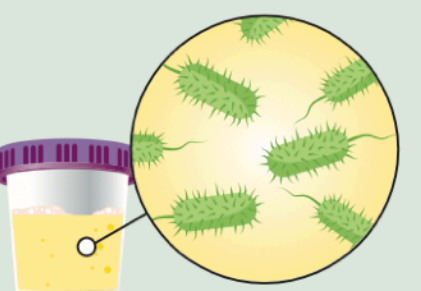
Systemic involvement

- Make a distinction between UTI with and without systemic involvement and avoid the term complicated UTI



Pyuria

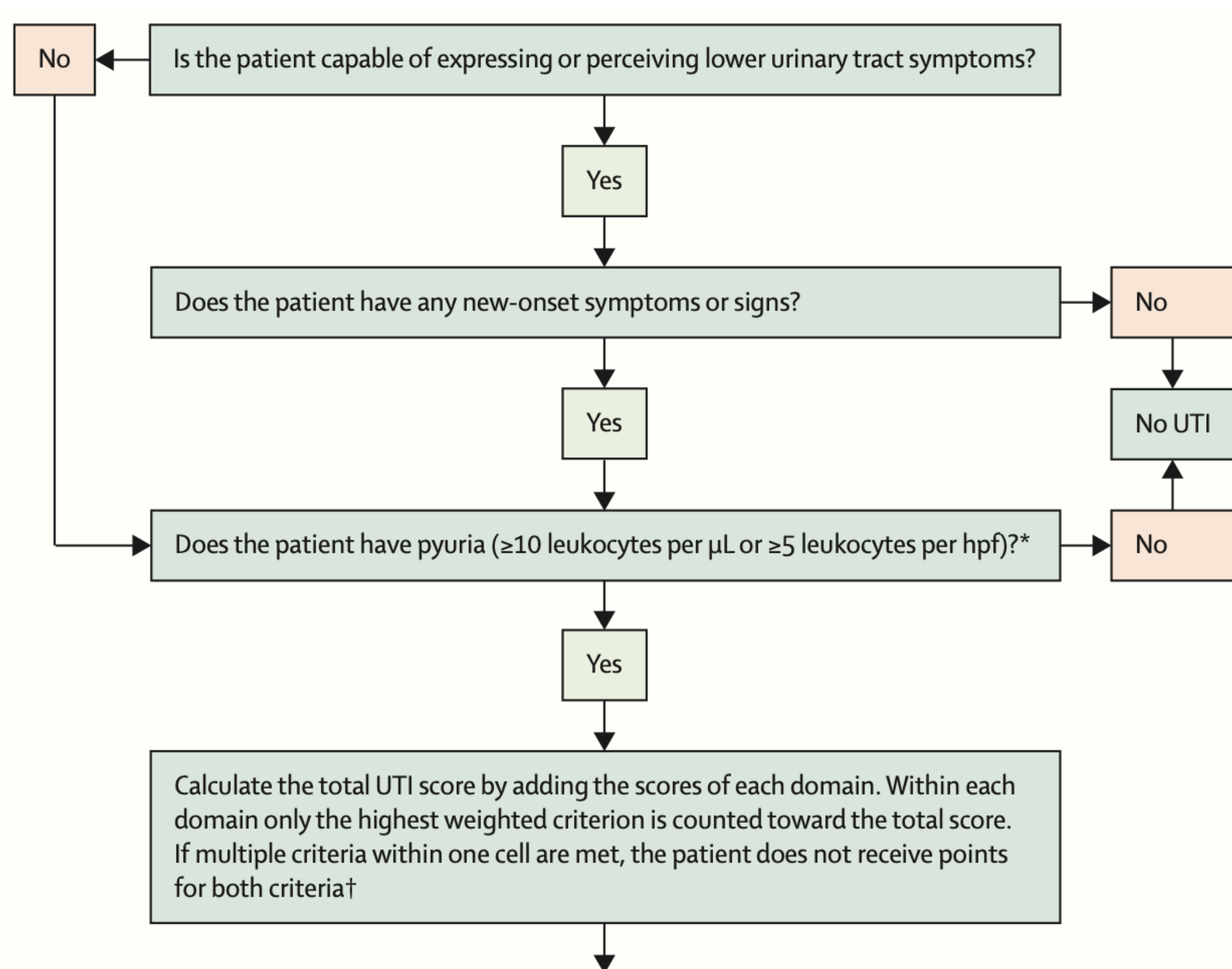
- Quantification of pyuria is encouraged, even if the study is conducted in non-hospital settings
- A new threshold of 200 leukocytes per μ L is introduced to identify clinically significant pyuria



Culture results

- Obtaining a urine culture is encouraged, even if the study is conducted in non-hospital settings
- Apply a threshold of 10⁴ CFU/mL for midstream urine samples and 10³ CFU/mL if *Escherichia coli* is cultured or if the sample was obtained through single in-out catheterisation

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Domain	Specifics	3 points	2 points	1 point	0 points
Symptoms and signs	Major Dysuria Urinary urgency Urinary frequency Minor Suprapubic pain or tenderness Flank pain or tenderness Perineal pain or painful prostate exam Urinary incontinence Macroscopic haematuria Recognition of symptoms	Two or more new-onset major symptoms Add 1 point if woman aged ≥65 years	One new-onset major symptom or two new-onset minor symptoms	One new-onset minor symptom or inability to express or experience symptoms (eg, delirium, dementia, or spinal cord injury)	
Systemic criteria	Fever (≥38.0°C or 100.4°F)‡ C-reactive protein concentration ≥50 mg/L‡ Procalcitonin concentration ≥0.5 ng/mL‡ White blood cell count ≥12×10 ⁹ per L‡ Bacteraemia (same pathogen as urine culture)		One or more new-onset systemic criteria		No systemic criteria
Pyuria	In either midstream urine or single catheterised sample	≥200 leukocytes per µl or ≥50 leukocytes per hpf	10–200 leukocytes per µl or 5–50 leukocytes per hpf Subtract 1 point if woman aged ≥65 years		
Culture results	Uropathogens: Enterobacterales, <i>Pseudomonas aeruginosa</i> , and <i>Staphylococcus saprophyticus</i>	One or two uropathogens: for midstream urine ≥10 ⁴ CFU/mL, for <i>Escherichia coli</i> ≥10 ³ CFU/mL is sufficient; for single catheter ≥10 ³ CFU/mL Subtract 1 point if woman aged ≥65 years Or Bacteraemia (same pathogen as urine culture)		Mixed flora or Other pathogen or No growth, but pretreated with antimicrobials	No growth, not pretreated with antimicrobials or Urine culture not performed

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Add domain scores
(symptoms + systemic criteria + pyuria + culture)

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Total UTI score
Definite UTI ≥8 points
Probable UTI 5–7 points
Possible UTI 3–4 points
No UTI 0–2 points

- UTI with systemic involvement if ≥1 criterion**
- Fever without other cause
 - Flank pain or costovertebral angle tenderness
 - Systolic blood pressure <90 mm Hg without other cause
 - Elevated inflammatory levels without other cause
 - Bacteraemia (same pathogen as urine culture)
 - Renal abscess
 - Pyonephrosis

Contact us for assistance:

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